benchmarking tools

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Contents

1 ENV

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutraceReproducibility/Figure2/")
```

2 Libraries

```r
library(ggsci)
## Warning: package 'ggsci' was built under R version 4.3.3
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##     filter, lag
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
library(scales)
library(readxl)
library(forcats)
library(ggplot2)
library(reshape2)
library(pheatmap)
library(circlize)
## ========================================
## circlize version 0.4.16
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
##   in R. Bioinformatics 2014.
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(circlize))
## ========================================
library(patchwork)
library(gridExtra)
##
## Attaching package: 'gridExtra'
## The following object is masked from 'package:dplyr':
##
##     combine
library(bedtoolsr)
## Warning in fun(libname, pkgname): bedtoolsr was built with bedtools version 2.30.0 but you have vers
## options(bedtools.path = \"[bedtools path]\")
library(ComplexHeatmap)
## Loading required package: grid
## ========================================
## ComplexHeatmap version 2.18.0
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##     genomic data. Bioinformatics 2016.
```

```
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## ========================================
## ! pheatmap() has been masked by ComplexHeatmap::pheatmap(). Most of the arguments
##    in the original pheatmap() are identically supported in the new function. You
##    can still use the original function by explicitly calling pheatmap::pheatmap().
##
## Attaching package: 'ComplexHeatmap'
## The following object is masked from 'package:pheatmap':
##
##     pheatmap
```

3 Setting

```
sample_size <- 0.6
iteration_times <- 50
```

4 loadData ENCODE

ENCDO451RUA transverse colon tissue Info: https://www.encodeproject.org/matrix/?type=Experiment&searchTerm=ENCDO451RUA&assay_title=snATAC-seq&assay_title=WGS snATAC (ENCFF491HQL): https://www.encodeproject.org/experiments/ENCSR349XKD/ WGS (ENCFF907ASL): https://www.encodeproject.org/experiments/ENCSR613XEH/

ENCDO845WKR transverse colon tissue Info: https://www.encodeproject.org/matrix/?type=Experiment&searchTerm=ENCDO845WKR&assay_title=snATAC-seq&assay_title=WGS snATAC (ENCFF354SCV): https://www.encodeproject.org/experiments/ENCSR434SXE/ WGS (ENCFF501WAL): https://www.encodeproject.org/experiments/ENCSR410ALS/

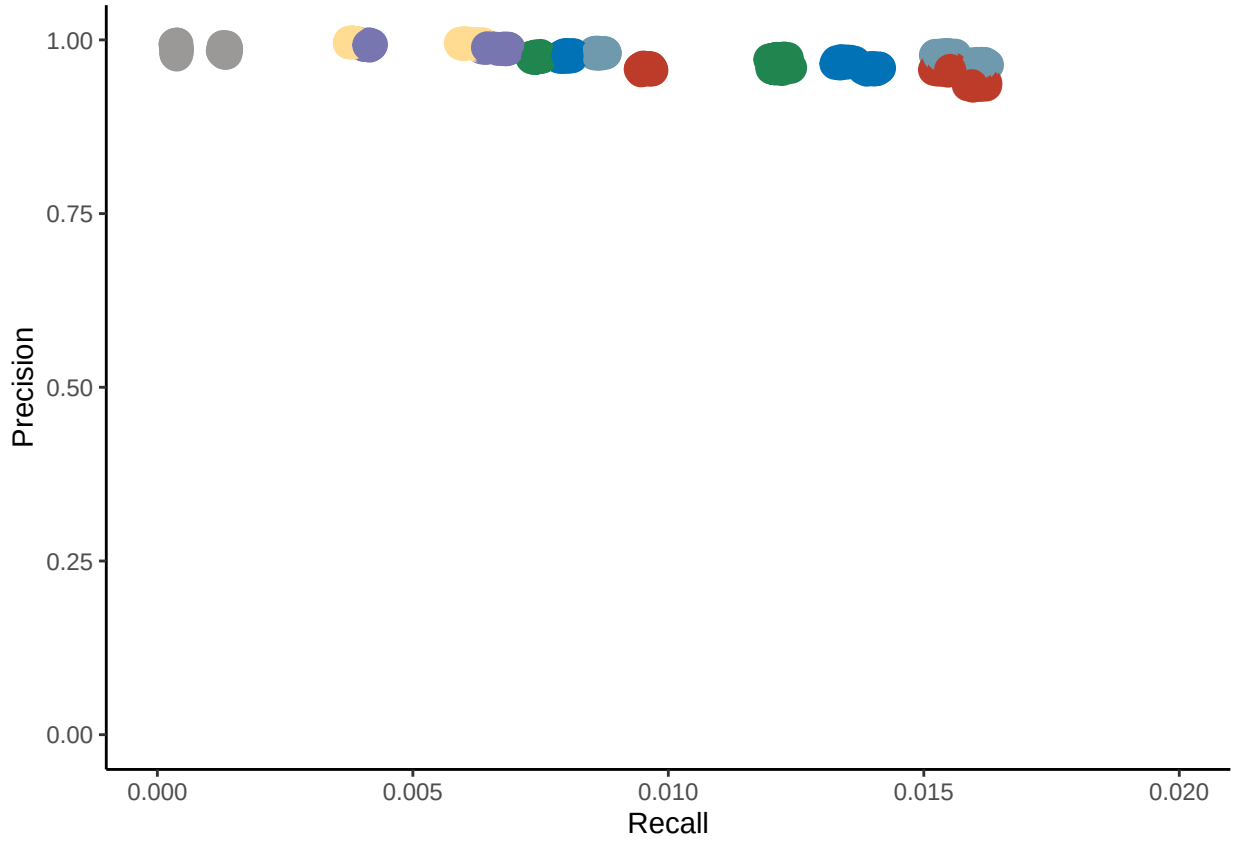
ENCDO793LXB transverse colon tissue Info: https://www.encodeproject.org/matrix/?type=Experiment&searchTerm=ENCDO793LXB&assay_title=snATAC-seq&assay_title=WGS snATAC (ENCFF692LTO): https://www.encodeproject.org/experiments/ENCSR997YNO/ WGS (ENCFF926UDZ): https://www.encodeproject.org/experiments/ENCSR997HAI/

```
path_encode_excel <- "../Data/GermlineMutation_results_HumanColon_sup.xlsx"
excel_sheets(path_encode_excel)
## [1] "freebayes" "Monopogen" "SAMtools"  "scAllele"  "scMutrace" "Strelka2"
## [7] "VarScan2"

# init res
col_benchmarking <- c("sampleID", "Iteration", "Tool", "Precision", "Recall", "F1Score")
SampleIDs_HumanColon <- c("ENCDO451RUA", "ENCDO793LXB", "ENCDO845WKR")
Tools_benchmarking_germline <- c("freebayes", "Monopogen", "SAMtools", "scAllele",
                         "scMutrace", "Strelka2", "VarScan2")
data_benchmarking_humanColon_res <- as.data.frame(matrix(data = 0, nrow = length(SampleIDs_HumanColon) *
```

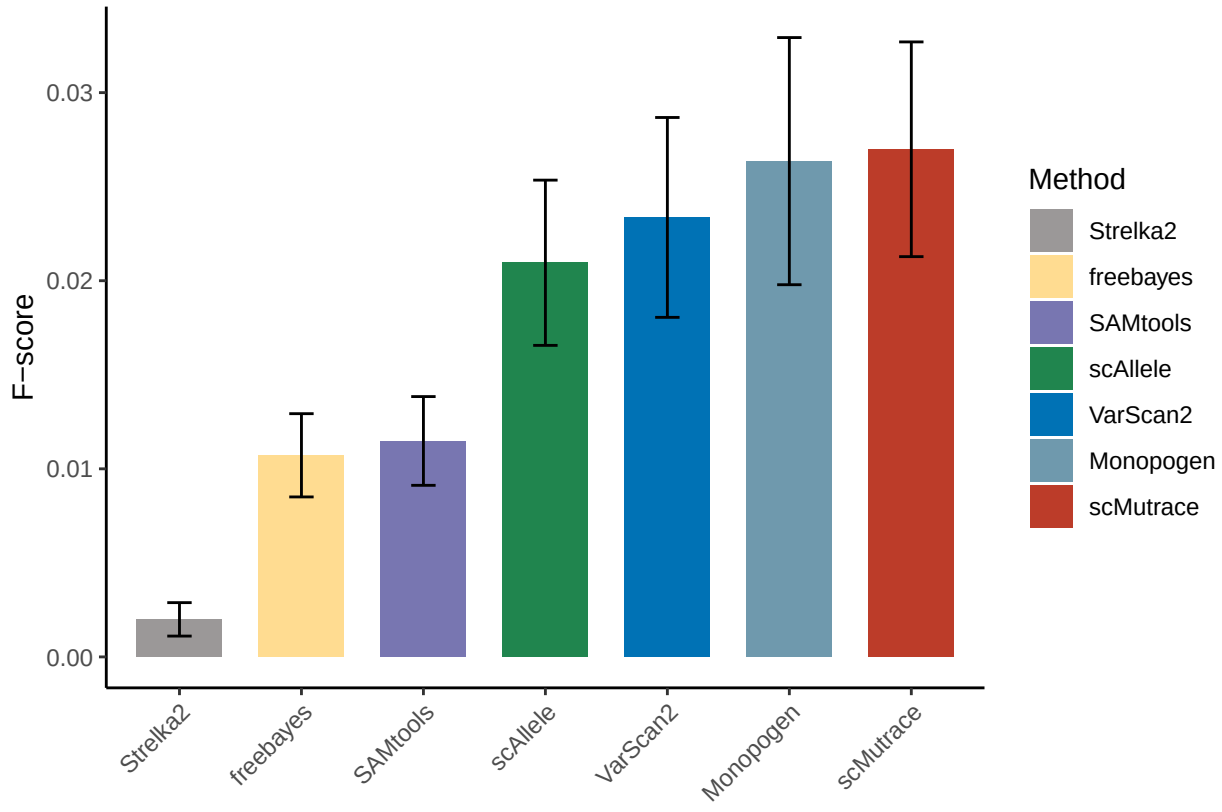
```r
colnames(data_benchmarking_humanColon_res) <- col_benchmarking

# generate benchmarking data
HumanColon_WGS <- readRDS("../Data/GermlineMutation_results_HumanColon_WGS.rds")
HumanColon_commonBed <- readRDS("../Data/GermlineMutation_results_HumanColon_commonBed.rds")

HumanColon_WGS_benchmarking <- list()
for (sampleID in SampleIDs_HumanColon){
  #data_bed <- HumanColon_commonBed[[sampleID]]
  #data_wgs <- HumanColon_WGS[[sampleID]]
  #data_wgs_benchmarking <- bedtoolsr::bt.intersect(data_wgs, data_bed)
  #colnames(data_wgs_benchmarking) <- c("chrom_wgs", "pos_start_wgs", "pos_end_wgs", "ref_wgs", "alt_wg
  #HumanColon_WGS_benchmarking[[sampleID]] <- data_wgs_benchmarking
  data_wgs <- HumanColon_WGS[[sampleID]]
  colnames(data_wgs) <- c("chrom_wgs", "pos_start_wgs", "pos_end_wgs", "ref_wgs", "alt_wgs" )
  HumanColon_WGS_benchmarking[[sampleID]] <- data_wgs
}

HumanColon_RNA_benchmarking <- list()
for (sampleID in SampleIDs_HumanColon){
  data_rna_ini <- data.frame()
  for (tool in Tools_benchmarking_germline){
    #data_rna_total <- read_excel(path_encode_excel, sheet = tool) %>% filter(SampleID == sampleID)
    #data_rna_sample_benchmarking <- bedtoolsr::bt.intersect(data_rna_total, HumanColon_commonBed[[samp
    #data_rna_ini <- rbind(data_rna_ini, data_rna_sample_benchmarking)

    data_rna_total <- read_excel(path_encode_excel, sheet = tool) %>% filter(SampleID == sampleID)
    data_rna_ini <- rbind(data_rna_ini, data_rna_total)
  }
  colnames(data_rna_ini) <- c("chrom_rna", "pos_start_rna", "pos_end_rna",
                              "ref_rna", "alt_rna", "Type_rna", "Method_rna",
                              "SampleID",   "Genome",   "SequencingType_rna")
  HumanColon_RNA_benchmarking[[sampleID]] <- data_rna_ini
}

HumanColon_Data_benchmarking <- list()
for (sampleID in SampleIDs_HumanColon){
  data_RNA_WGS <- data.frame()
  data_WGS <- HumanColon_WGS_benchmarking[[sampleID]]
  data_WGS$id <- paste(data_WGS$chrom_wgs, data_WGS$pos_start_wgs, data_WGS$ref_wgs, sep = "_")
  for (tool in Tools_benchmarking_germline){
    data_RNA <- HumanColon_RNA_benchmarking[[sampleID]] %>% filter(Method_rna == tool)
    data_RNA$id <- paste(data_RNA$chrom_rna, data_RNA$pos_start_rna, data_RNA$ref_rna, sep = "_")
    data_RNA_WGS_tool <- merge(data_RNA, data_WGS, by = "id", all = T)
    data_RNA_WGS_tool$Method_rna <- tool
    data_RNA_WGS <- rbind(data_RNA_WGS, data_RNA_WGS_tool)
  }
  HumanColon_Data_benchmarking[[sampleID]] <- data_RNA_WGS
}

# bootstrap
set.seed(2025)
line_num <- 0
```

```r
for (iter in 1:iteration_times){
  for (sampleID in SampleIDs_HumanColon){
    HumanColon_Data_benchmarking_bm <- HumanColon_Data_benchmarking[[sampleID]] %>% sample_frac(sample_s
    for (tool in Tools_benchmarking_germline){
      line_num <- line_num + 1
      HumanColon_Data_benchmarking_bm_tool <- HumanColon_Data_benchmarking_bm %>% filter(Method_rna == t
      total_calls <- sum(!is.na(HumanColon_Data_benchmarking_bm_tool$chrom_rna))
      total_truth <- sum(!is.na(HumanColon_Data_benchmarking_bm_tool$chrom_wgs))
      tp <- nrow(na.omit(HumanColon_Data_benchmarking_bm_tool))
      fp <- total_calls - tp
      fn <- total_truth - tp
      sensitivity <- ifelse(total_truth > 0, tp / (tp + fn), NA)
      precision   <- ifelse(total_calls > 0, tp / (tp + fp), NA)
      f1          <- ifelse(!is.na(sensitivity) & !is.na(precision) &
                            (sensitivity + precision) > 0,
                            2 * (precision * sensitivity) / (precision + sensitivity),
                            NA)
      data_benchmarking_humanColon_res[line_num, "sampleID"] <- sampleID
      data_benchmarking_humanColon_res[line_num, "Iteration"] <- iter
      data_benchmarking_humanColon_res[line_num, "Tool"] <- tool
      data_benchmarking_humanColon_res[line_num, "Precision"] <- precision
      data_benchmarking_humanColon_res[line_num, "Recall"] <- sensitivity
      data_benchmarking_humanColon_res[line_num, "F1Score"] <- f1
    }
  }
}

colorAssign_0 = c(freebayes = "#FFDC91FF", SAMtools = "#7876B1FF",
                  scAllele = "#20854EFF", VarScan2 = "#0072B5FF",
                  Strelka2= "#9B9898", Monopogen = "#6F99ADFF",
                  scMutrace = "#BC3C29FF")

data_benchmarking_humanColon_res <- data_benchmarking_humanColon_res %>%
  mutate(Tool = fct_reorder(Tool, F1Score, .fun = mean, .na_rm = TRUE))

#data_benchmarking_humanColon_res$Tool <- factor(data_benchmarking_humanColon_res$Tool, levels = c("Str

data_benchmarking_humanColon_res <- na.omit(data_benchmarking_humanColon_res)

p_humanColon_point <- ggplot(data = data_benchmarking_humanColon_res) +
  geom_point(mapping =  aes(x = Recall, y = Precision, color = Tool), size = 5, show.legend = FALSE) +
  scale_color_manual(values = colorAssign_0) +
  xlab("Recall") + ylab("Precision") +
  xlim(0, 0.02) +
  ylim(0, 1) +
  theme_classic()

p_humanColon_box <- ggplot(data = data_benchmarking_humanColon_res, aes(x = Tool, y = F1Score)) +
  stat_summary(mapping=aes(fill = Tool),fun=mean, geom = "bar", fun.args = list(mult=1),width=0.7) +
  stat_summary(fun.data=mean_sdl,fun.args = list(mult=1), geom="errorbar",width=0.2) +
  scale_fill_manual(values = colorAssign_0) +
  labs(x = "",y = "F-score")  +
  scale_x_discrete(guide = guide_axis(angle = 45)) +
```

```
  theme_classic() + guides(fill=guide_legend(title="Method"))

p_humanColon_point
```




```
p_humanColon_box
```



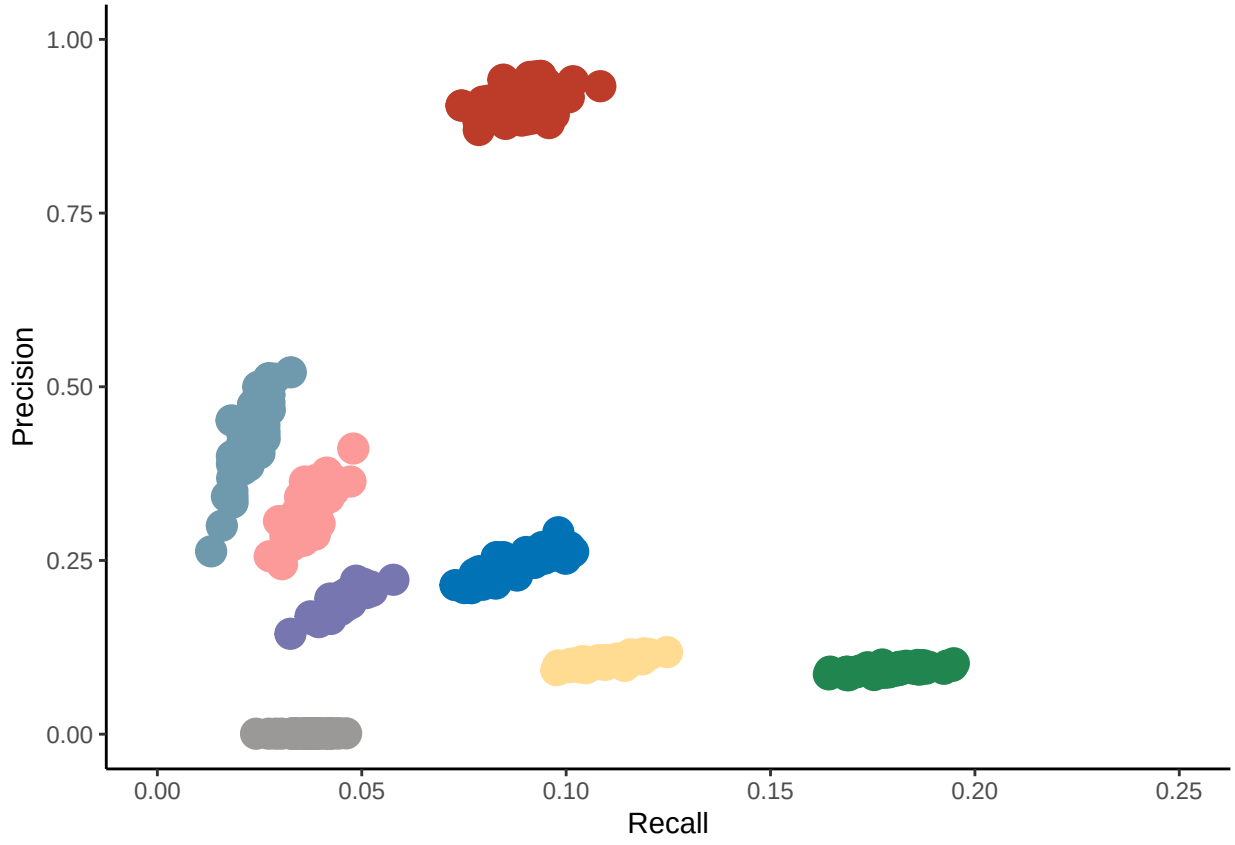

```
ggsave("../Figures/p_humanColon_benchmarking_point.pdf", p_humanColon_point, width = 4.47, height = 3.75
ggsave("../Figures/p_humanColon_benchmarking_box.pdf", p_humanColon_box, width = 4.47, height = 3.75)
```

5 cSCC

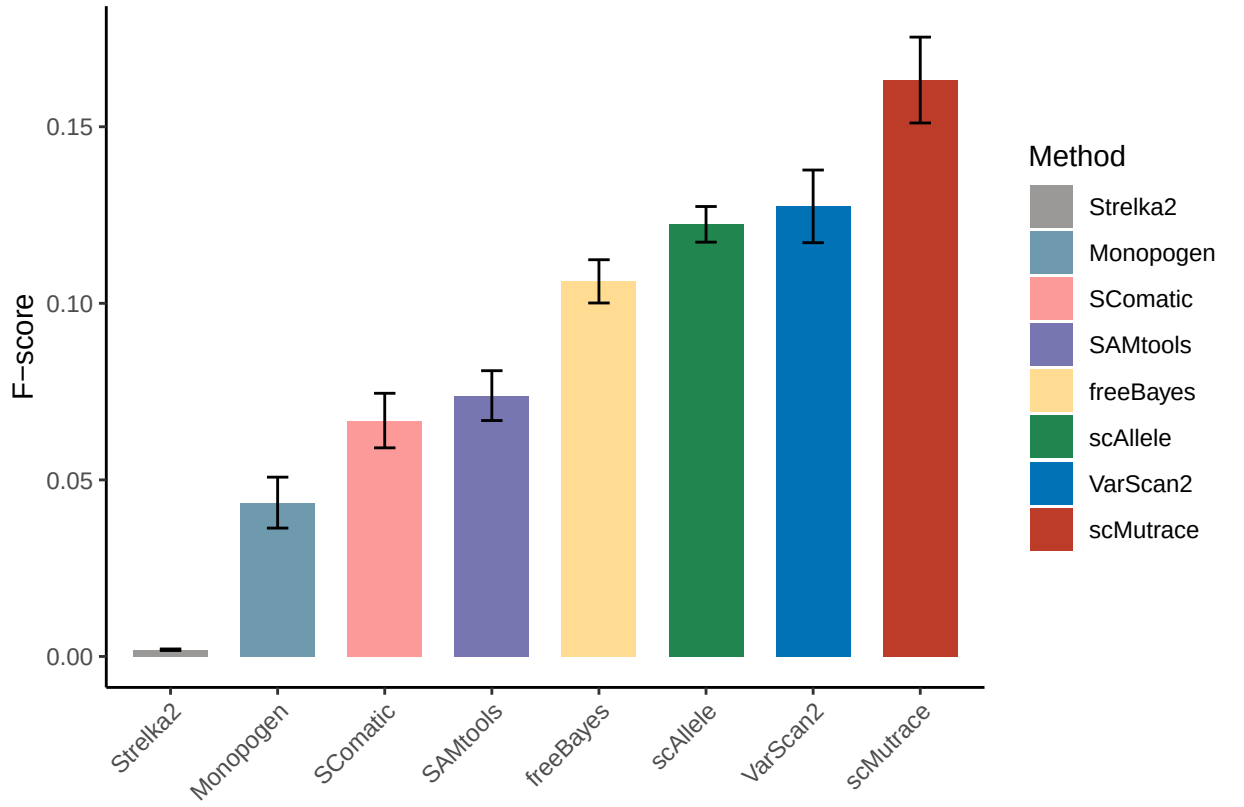
```
path_cSCC_excel <- "../Data/SomaticMutation_results_cSCC_sup.xlsx"
excel_sheets(path_cSCC_excel)
##  [1] "WES"       "freeBayes" "Monopogen" "SAMtools"  "scAllele"  "scMutrace"
##  [7] "SComatic"  "Strelka2"  "VarScan2"  "SBS"

Tools_benchmarking_cSCC <- c("freeBayes", "Monopogen", "SAMtools", "scAllele","scMutrace", "SComatic",

SampleIDs_cSCC <- c("P2", "P3", "P4", "P5", "P6", "P7", "P8", "P9", "P10")

col_benchmarking_cSCC <- c("Iteration", "Tool", "Precision", "Recall", "F1Score")

data_benchmarking_cSCC_res <- as.data.frame(matrix(data = 0, nrow = iteration_times * length(Tools_bench
                                                   ncol = length(col_benchmarking_cSCC)))

colnames(data_benchmarking_cSCC_res) <- col_benchmarking_cSCC

cSCC_WES <- read_excel(path_cSCC_excel, sheet = "WES")
```

```
cSCC_WES$id <- paste(cSCC_WES$chrom, cSCC_WES$pos_start, cSCC_WES$ref, cSCC_WES$SampleID, sep = "_")
colnames(cSCC_WES) <- c("Project_WES", "Sample_WES", "ID_WES", "Genome_WES",
                        "mut_type_WES",  "chrom_WES", "pos_start_WES", "pos_end_WES",
                        "ref_WES", "alt_WES", "Type_WES", "Method_WES", "SampleID_WES",
                        "id" )

cSCC_Data_benchmarking <- list()

for (tool in Tools_benchmarking_cSCC){
  cSCC_RNA <- read_excel(path_cSCC_excel, sheet = tool)
  cSCC_RNA$id <- paste(cSCC_RNA$chrom, cSCC_RNA$pos_start, cSCC_RNA$ref, cSCC_RNA$SampleID, sep = "_")
  colnames(cSCC_RNA) <- c("Project_RNA", "Sample_RNA", "ID_RNA", "Genome_RNA",
                        "mut_type_RNA",  "chrom_RNA", "pos_start_RNA", "pos_end_RNA",
                        "ref_RNA", "alt_RNA", "Type_RNA", "Method_RNA", "SampleID_RNA",
                        "id" )
  cSCC_WES_RNA_tool <- merge(cSCC_WES, cSCC_RNA, by = "id", all = T)
  cSCC_WES_RNA_tool$Method_RNA <- tool
  cSCC_Data_benchmarking[[tool]] <- cSCC_WES_RNA_tool
}


set.seed(2025)
line_num <- 0
for (iter in 1:iteration_times){
  for (tool in Tools_benchmarking_cSCC){
    line_num <- line_num + 1
    cSCC_Data_benchmarking_bm <- cSCC_Data_benchmarking[[tool]] %>% sample_frac(sample_size)
    cSCC_Data_benchmarking_bm_tool <- cSCC_Data_benchmarking_bm %>% filter(Method_RNA == tool)
    total_calls <- sum(!is.na(cSCC_Data_benchmarking_bm_tool$chrom_RNA))
    total_truth <- sum(!is.na(cSCC_Data_benchmarking_bm$chrom_WES))
    tp <- nrow(na.omit(cSCC_Data_benchmarking_bm_tool))
    fp <- total_calls - tp
    fn <- total_truth - tp
    sensitivity <- ifelse(total_truth > 0, tp / (tp + fn), NA)
    precision   <- ifelse(total_calls > 0, tp / (tp + fp), NA)
    f1          <- ifelse(!is.na(sensitivity) & !is.na(precision) &
                          (sensitivity + precision) > 0,
                          2 * (precision * sensitivity) / (precision + sensitivity),
                          NA)
    data_benchmarking_cSCC_res[line_num, "Iteration"] <- iter
    data_benchmarking_cSCC_res[line_num, "Tool"] <- tool
    data_benchmarking_cSCC_res[line_num, "Precision"] <- precision
    data_benchmarking_cSCC_res[line_num, "Recall"] <- sensitivity
    data_benchmarking_cSCC_res[line_num, "F1Score"] <- f1
  }
}

colorAssign_1 = c(freeBayes = "#FFDC91FF", SAMtools = "#7876B1FF",
                  scAllele = "#20854EFF", VarScan2 = "#0072B5FF",
                  Strelka2= "#9B9898", SComatic = "#fb9a99",
                  Monopogen = "#6F99ADFF", scMutrace = "#BC3C29FF")

data_benchmarking_cSCC_res <- data_benchmarking_cSCC_res %>%
```

```
  mutate(Tool = fct_reorder(Tool, F1Score, .fun = mean, .na_rm = TRUE))

data_benchmarking_cSCC_res <- na.omit(data_benchmarking_cSCC_res)

# c("freeBayes", "Monopogen", "SAMtools", "scAllele","scMutrace", "SComatic", "Strelka2", "VarScan2")
# data_benchmarking_cSCC_res$Tool <- factor(data_benchmarking_cSCC_res$Tool, levels = c("Strelka2", "fr

p_cSCC_point <- ggplot(data = data_benchmarking_cSCC_res) +
  geom_point(mapping =  aes(x = Recall, y = Precision, color = Tool), size = 5, show.legend = FALSE) +
  scale_color_manual(values = colorAssign_1) +
  xlab("Recall") + ylab("Precision") +
  xlim(0, 0.25) +
  ylim(0, 1) +
  theme_classic()

p_cSCC_box <- ggplot(data = data_benchmarking_cSCC_res, aes(x = Tool, y = F1Score)) +
  stat_summary(mapping=aes(fill = Tool),fun=mean, geom = "bar", fun.args = list(mult=1),width=0.7) +
  stat_summary(fun.data=mean_sdl,fun.args = list(mult=1), geom="errorbar",width=0.2) +
  scale_fill_manual(values = colorAssign_1) +
  labs(x = "",y = "F-score")  +
  scale_x_discrete(guide = guide_axis(angle = 45)) +
  theme_classic() + guides(fill=guide_legend(title="Method"))

p_cSCC_point
```



```
p_cSCC_box
```

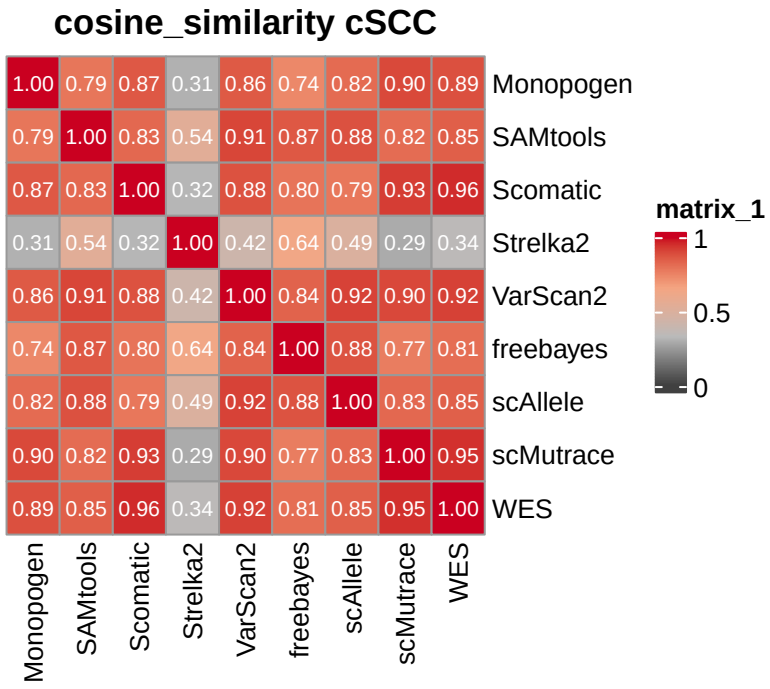



```
ggsave("../Figures/p_cSCC_benchmarking_point.pdf", p_cSCC_point, width = 4.47, height = 3.75)
ggsave("../Figures/p_cSCC_benchmarking_box.pdf", p_cSCC_box, width = 4.47, height = 3.75)
```

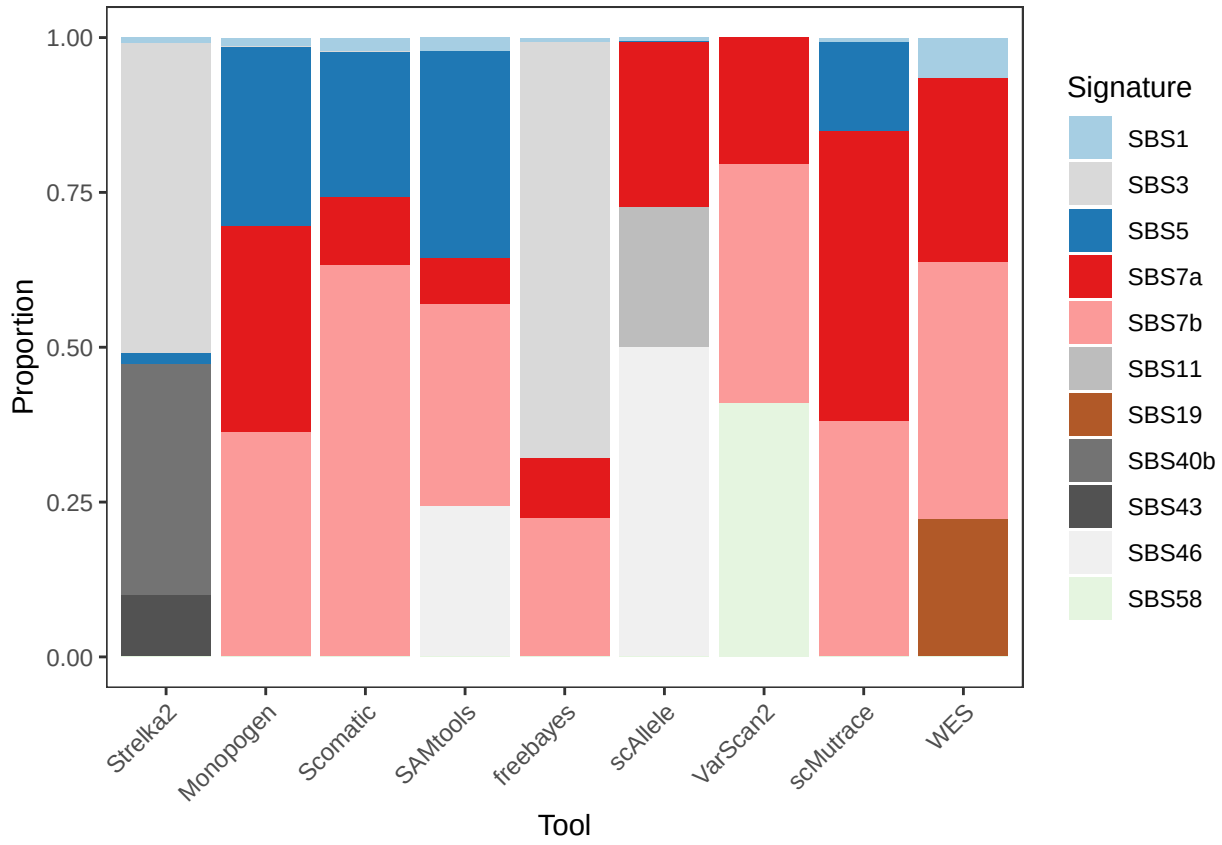
6 cSCC mutation signature

```
cosine_similarity <- function(a, b) {
  sum(a * b) / (sqrt(sum(a^2)) * sqrt(sum(b^2)))
}

cSCC_sig <- read.table("../data/Mutational_Matrix_SBS_cSCC.txt", header = T, sep = "\t")
cSCC_contribution_raw <- read.csv("../data/Signature_contributions_cSCC.csv", header = T, row.names = 1)
cSCC_contribution <- cSCC_contribution_raw[,-1]

# calsim
cs_Monopogen_WES <- cosine_similarity(cSCC_sig$Monopogen, cSCC_sig$WES)
cs_SAMtools_WES <- cosine_similarity(cSCC_sig$SAMtools, cSCC_sig$WES)
cs_Scomatic_WES <- cosine_similarity(cSCC_sig$Scomatic, cSCC_sig$WES)
cs_Strelka2_WES <- cosine_similarity(cSCC_sig$Strelka2, cSCC_sig$WES)
cs_VarScan2_WES <- cosine_similarity(cSCC_sig$VarScan2, cSCC_sig$WES)
cs_freebayes_WES <- cosine_similarity(cSCC_sig$freebayes, cSCC_sig$WES)
```

```r
cs_scAllele_WES <- cosine_similarity(cSCC_sig$scAllele, cSCC_sig$WES)
cs_scMutrace_WES <- cosine_similarity(cSCC_sig$scMutrace, cSCC_sig$WES)

cSCC_cossim_tools <- c("Monopogen", "SAMtools", "Scomatic", "Strelka2", "VarScan2",
                       "freebayes", "scAllele", "scMutrace", "WES")

cSCC_data_cosSim_res <- as.data.frame(matrix(data = 0, nrow = length(cSCC_cossim_tools),
                                             ncol = length(cSCC_cossim_tools)))
rownames(cSCC_data_cosSim_res) <- cSCC_cossim_tools
colnames(cSCC_data_cosSim_res) <- cSCC_cossim_tools

for (i in 1:nrow(cSCC_data_cosSim_res)){
  row_item_tmp <- rownames(cSCC_data_cosSim_res)[i]
  for (j in 1:ncol(cSCC_data_cosSim_res)){
    col_item_tmp <- colnames(cSCC_data_cosSim_res)[j]
    score_tmp <- cosine_similarity(cSCC_sig[,row_item_tmp], cSCC_sig[,col_item_tmp])
    cSCC_data_cosSim_res[i,j] <- score_tmp
  }
}

col_fun_v1 <- colorRamp2(c(0, 0.33,  0.66, 1), c("#404040", "#bababa", "#f4a582", "#ca0020"))

p_cSCC_data_cosSim <- pheatmap(cSCC_data_cosSim_res,
                               cellwidth = 20, cellheight = 20,
                               cluster_rows = F, cluster_cols = F,
                               display_numbers = TRUE, number_color = "white",
                               main = "cosine_similarity cSCC",
                               color = col_fun_v1)
## Warning: The input is a data frame, convert it to the matrix.
p_cSCC_data_cosSim
```




```r
cSCC_contribution_long <- melt(as.matrix(cSCC_contribution))
colnames(cSCC_contribution_long) <- c("Signature", "Tool", "Value")

sbs_color <- c(
  "SBS1" = "#a6cee3", "SBS3" = "#d9d9d9", "SBS5" = "#1f78b4",
  "SBS7a" = "#e31a1c", "SBS7b" = "#fb9a99", "SBS11" = "#bdbdbd",
  "SBS19" = "#b15928", "SBS40b" = "#737373", "SBS43" = "#525252",
  "SBS46" = "#f0f0f0", "SBS58" = "#e5f5e0"
)

cSCC_contribution_long$Tool <- factor(cSCC_contribution_long$Tool, levels = c("Strelka2", "Monopogen",
                                                                              "scAllele", "VarScan2", "

p_cSCC_contribution <- ggplot(cSCC_contribution_long, aes(x = Tool, y = Value, fill = Signature)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = sbs_color) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid = element_blank()
  ) +
  labs(x = "Tool", y = "Proportion", fill = "Signature")

p_cSCC_contribution
```



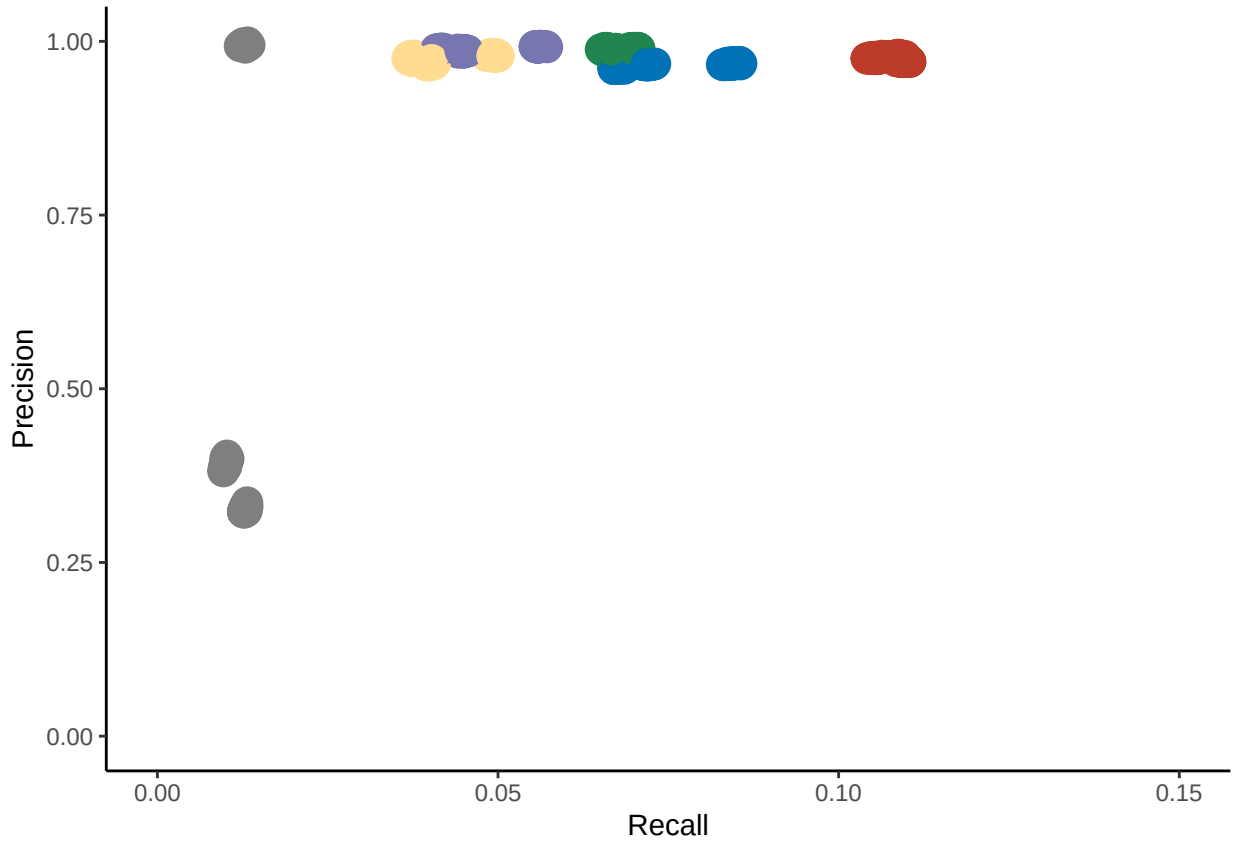

```
pdf("../Figures/p_cSCC_data_cosSim.pdf", width = 5.31, height = 3.90)
p_cSCC_data_cosSim
dev.off()
## pdf
##   2
ggsave("../Figures/p_cSCC_contribution.pdf", p_cSCC_contribution, width = 4.47, height = 3.75)
```

7 Mouse scATAC

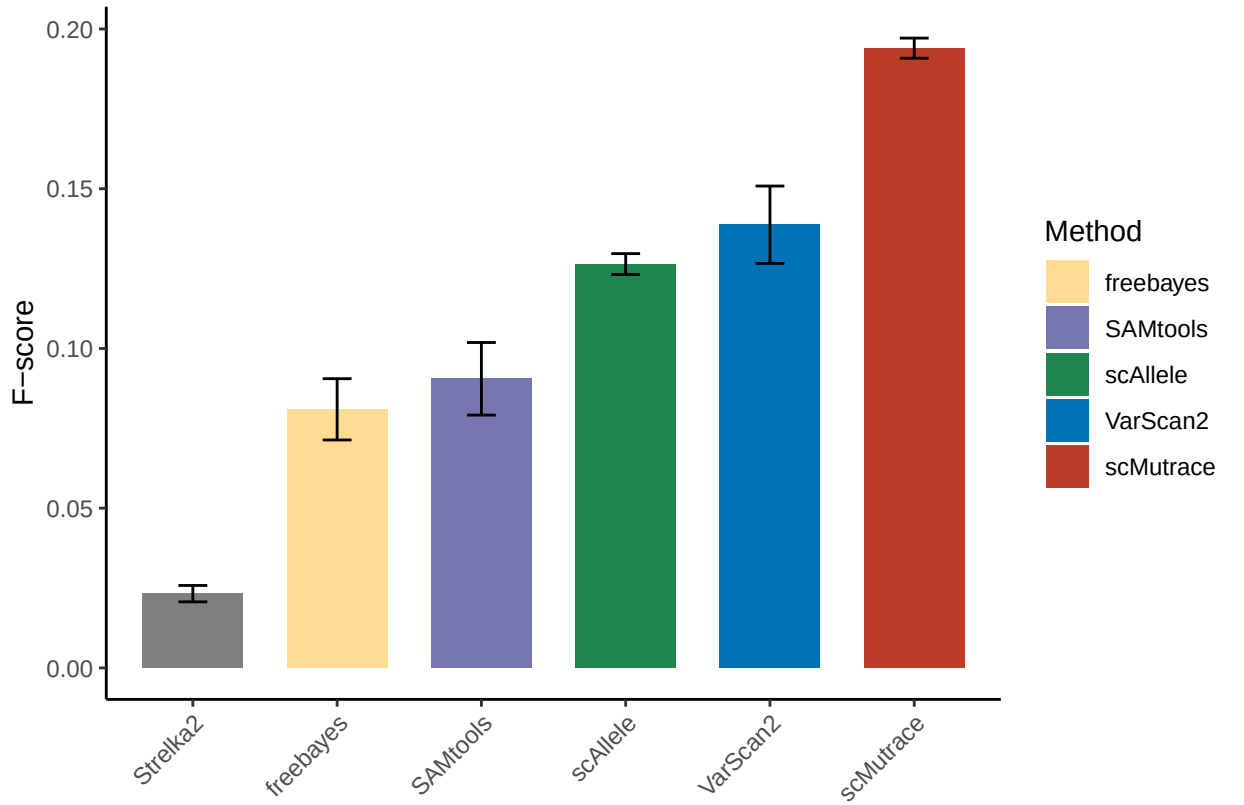
```
Tools_benchmarking_Mouse <- c("freebayes", "SAMtools", "scAllele",
                              "scMutrace", "Strelka2", "VarScan2")

SampleIDs_Mouse <- c("EPSRC1", "EPSRC2", "EPSRC4")

#Mouse_WGS <- readRDS("../Data/GermlineMutation_results_Mouse_WGS_scATAC.rds")
#Mouse_scATA <- readRDS("../Data/GermlineMutation_results_Mouse_scATAC.rds")

Mouse_WGS <- readRDS("../Data/GermlineMutation_results_Mouse_WGS_scATAC.rds")
Mouse_scATA <- readRDS("../Data/GermlineMutation_results_Mouse_scATAC.rds")

col_benchmarking_Mouse <- c("Iteration", "sampleID", "Tool", "Precision", "Recall", "F1Score")
```

```r
data_benchmarking_Mouse_scATA_res <- as.data.frame(matrix(data = 0, nrow = iteration_times * length(Tool
                                                  ncol = length(col_benchmarking_Mouse)))

colnames(data_benchmarking_Mouse_scATA_res) <- col_benchmarking_Mouse

Mouse_Data_scATAC_benchmarking <- list()

for (sampleID in SampleIDs_Mouse){
  mouse_wgs_sample_ini <- data.frame()
  mouse_wgs_sample <- Mouse_WGS[[sampleID]]
  mouse_wgs_sample$id <- paste(mouse_wgs_sample$chrom, mouse_wgs_sample$pos_start,
                               mouse_wgs_sample$ref, sep = "_")
  colnames(mouse_wgs_sample) <- c("chrom_wgs", "pos_start_wgs", "pos_end_wgs",
                                  "ref_wgs", "end_wgs", "Tool_wgs", "id")
  for (tool in Tools_benchmarking_Mouse){
    mouse_ata_sample_tool <- Mouse_scATA[[sampleID]] %>% filter(Tool == tool)
    mouse_ata_sample_tool$id <- paste(mouse_ata_sample_tool$chrom, mouse_ata_sample_tool$pos_start,
                                      mouse_ata_sample_tool$ref, sep = "_")
    colnames(mouse_ata_sample_tool) <- c("chrom_ata", "pos_start_ata", "pos_end_ata",
                                  "ref_ata", "end_ata", "Tool_ata", "id")

    mouse_wgs_ata_tmp <- merge(mouse_wgs_sample, mouse_ata_sample_tool, by = "id", all = T)
    mouse_wgs_ata_tmp$Tool_ata <- tool
    mouse_wgs_sample_ini <- rbind(mouse_wgs_sample_ini, mouse_wgs_ata_tmp)
  }
  Mouse_Data_scATAC_benchmarking[[sampleID]] <- mouse_wgs_sample_ini
}

set.seed(2025)
line_num <- 0
for (iter in 1:iteration_times){
  for (sampleID in SampleIDs_Mouse){
    Mouse_Data_scATAC_benchmarking_bm <- Mouse_Data_scATAC_benchmarking[[sampleID]] %>% sample_frac(samp
    for (tool in Tools_benchmarking_Mouse){
      line_num <- line_num + 1
      Mouse_Data_scATAC_benchmarking_bm_tool <- Mouse_Data_scATAC_benchmarking_bm %>% filter(Tool_ata ==
      total_calls <- sum(!is.na(Mouse_Data_scATAC_benchmarking_bm_tool$chrom_ata))
      total_truth <- sum(!is.na(Mouse_Data_scATAC_benchmarking_bm$chrom_wgs))
      tp <- nrow(na.omit(Mouse_Data_scATAC_benchmarking_bm_tool))
      fp <- total_calls - tp
      fn <- total_truth - tp
      sensitivity <- ifelse(total_truth > 0, tp / (tp + fn), NA)
      precision   <- ifelse(total_calls > 0, tp / (tp + fp), NA)
      f1          <- ifelse(!is.na(sensitivity) & !is.na(precision) &
                            (sensitivity + precision) > 0,
                            2 * (precision * sensitivity) / (precision + sensitivity),
                            NA)
      data_benchmarking_Mouse_scATA_res[line_num, "Iteration"] <- iter
      data_benchmarking_Mouse_scATA_res[line_num, "sampleID"] <- sampleID
      data_benchmarking_Mouse_scATA_res[line_num, "Tool"] <- tool
      data_benchmarking_Mouse_scATA_res[line_num, "Precision"] <- precision
      data_benchmarking_Mouse_scATA_res[line_num, "Recall"] <- sensitivity
      data_benchmarking_Mouse_scATA_res[line_num, "F1Score"] <- f1
```

```
    }
  }
}

colorAssign_1 = c(freebayes = "#FFDC91FF", SAMtools = "#7876B1FF",
                  scAllele = "#20854EFF", VarScan2 = "#0072B5FF",
                  strelka2= "#9B9898", SComatic = "#fb9a99",
                  Monopogen = "#6F99ADFF", scMutrace = "#BC3C29FF")

data_benchmarking_Mouse_scATA_res <- data_benchmarking_Mouse_scATA_res %>%
  mutate(Tool = fct_reorder(Tool, F1Score, .fun = mean, .na_rm = TRUE))

data_benchmarking_Mouse_scATA_res <- na.omit(data_benchmarking_Mouse_scATA_res)

p_Mouse_scATAC_point <- ggplot(data = data_benchmarking_Mouse_scATA_res) +
  geom_point(mapping =  aes(x = Recall, y = Precision, color = Tool), size = 5, show.legend = FALSE) +
  scale_color_manual(values = colorAssign_1) +
  xlab("Recall") + ylab("Precision") +
  xlim(0, 0.15) +
  ylim(0, 1) +
  theme_classic()

p_Mouse_scATAC_box <- ggplot(data = data_benchmarking_Mouse_scATA_res, aes(x = Tool, y = F1Score)) +
  stat_summary(mapping=aes(fill = Tool),fun=mean, geom = "bar", fun.args = list(mult=1),width=0.7) +
  stat_summary(fun.data=mean_sdl,fun.args = list(mult=1), geom="errorbar",width=0.2) +
  scale_fill_manual(values = colorAssign_1) +
  labs(x = "",y = "F-score")  +
  scale_x_discrete(guide = guide_axis(angle = 45)) +
  theme_classic() + guides(fill=guide_legend(title="Method"))

p_Mouse_scATAC_point
```




```
p_Mouse_scATAC_box
```



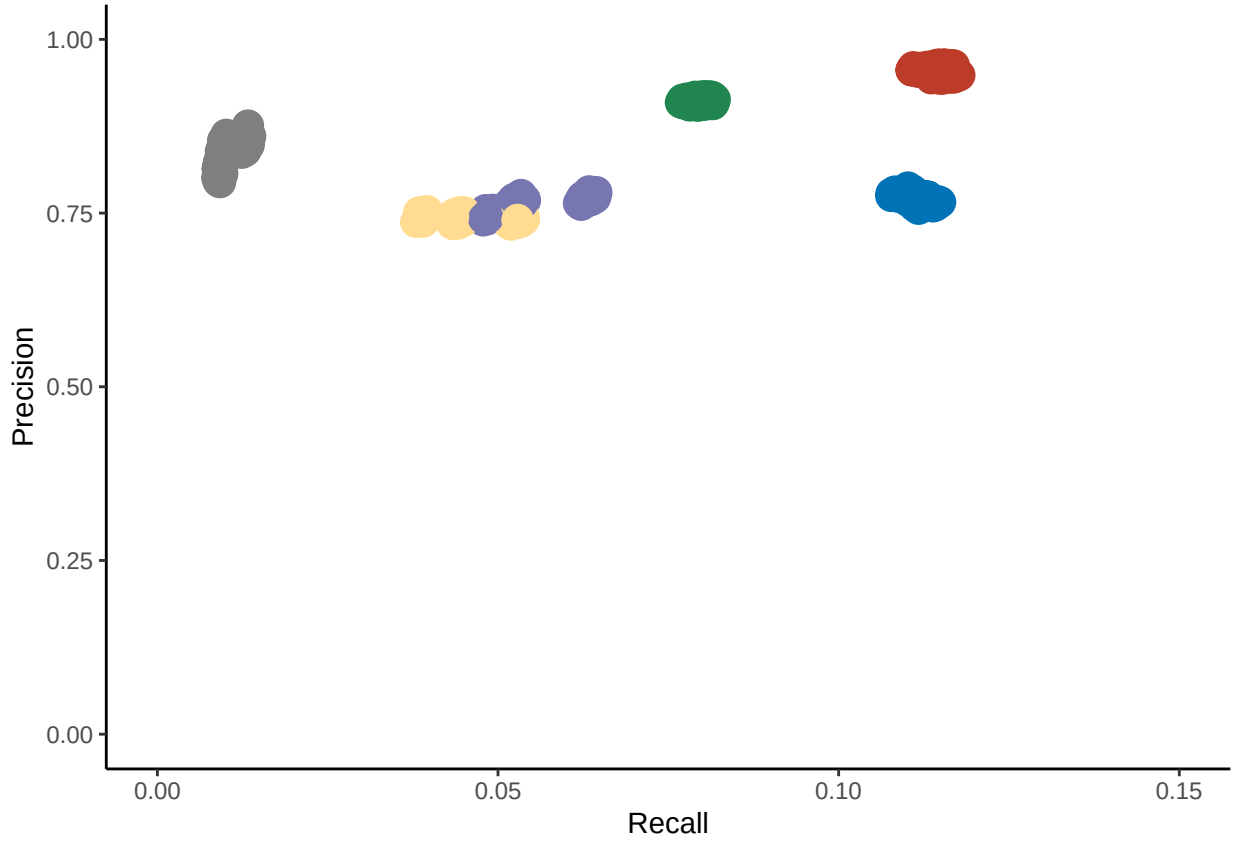

```
ggsave("../Figures/p_Mouse_scATAC_benchmarking_point.pdf", p_Mouse_scATAC_point, width = 4.47, height =
ggsave("../Figures/p_Mouse_scATAC_benchmarking_box.pdf", p_Mouse_scATAC_box, width = 4.47, height = 3.75
```

8 Mouse scRNA

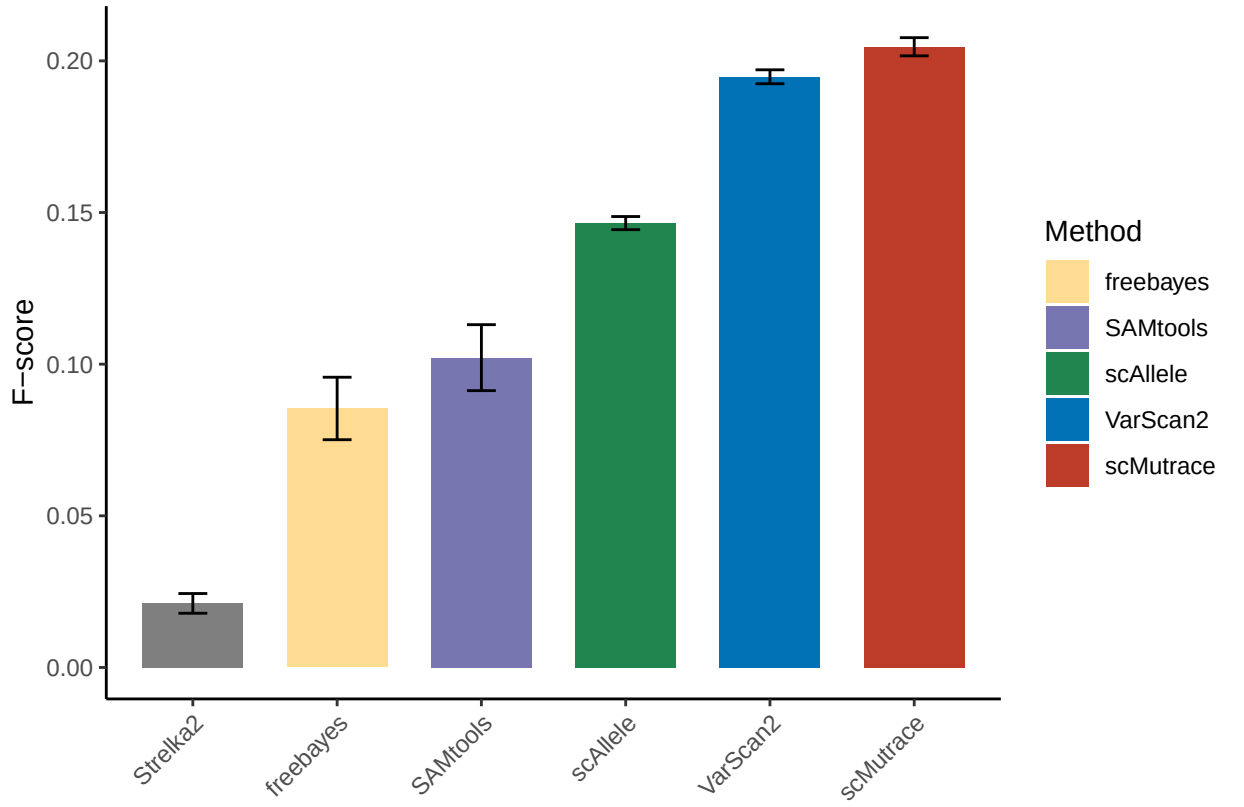
```
Tools_benchmarking_Mouse <- c("freebayes", "SAMtools", "scAllele",
                              "scMutrace", "Strelka2", "VarScan2")

SampleIDs_Mouse <- c("EPSRC1", "EPSRC2", "EPSRC4")

Mouse_WGS <- readRDS("../Data/GermlineMutation_results_Mouse_WGS_scRNA.rds")
Mouse_scRNA <- readRDS("../Data/GermlineMutation_results_Mouse_scRNA.rds")

col_benchmarking_Mouse <- c("Iteration", "sampleID", "Tool", "Precision", "Recall", "F1Score")

data_benchmarking_Mouse_scRNA_res <- as.data.frame(matrix(data = 0, nrow = iteration_times * length(Tool
                                                  ncol = length(col_benchmarking_Mouse)))

colnames(data_benchmarking_Mouse_scRNA_res) <- col_benchmarking_Mouse

Mouse_Data_scRNA_benchmarking <- list()
```

```r
for (sampleID in SampleIDs_Mouse){
  mouse_wgs_sample_ini <- data.frame()
  mouse_wgs_sample <- Mouse_WGS[[sampleID]]
  mouse_wgs_sample$id <- paste(mouse_wgs_sample$chrom, mouse_wgs_sample$pos_start,
                               mouse_wgs_sample$ref, sep = "_")
  colnames(mouse_wgs_sample) <- c("chrom_wgs", "pos_start_wgs", "pos_end_wgs",
                                  "ref_wgs", "end_wgs", "Tool_wgs", "id")
  for (tool in Tools_benchmarking_Mouse){
    mouse_rna_sample_tool <- Mouse_scRNA[[sampleID]] %>% filter(Tool == tool)
    mouse_rna_sample_tool$id <- paste(mouse_rna_sample_tool$chrom, mouse_rna_sample_tool$pos_start,
                                      mouse_rna_sample_tool$ref, sep = "_")
    colnames(mouse_rna_sample_tool) <- c("chrom_ata", "pos_start_ata", "pos_end_ata",
                                  "ref_ata", "end_ata", "Tool_ata", "id")

    mouse_wgs_rna_tmp <- merge(mouse_wgs_sample, mouse_rna_sample_tool, by = "id", all = T)
    mouse_wgs_rna_tmp$Tool_ata <- tool
    mouse_wgs_sample_ini <- rbind(mouse_wgs_sample_ini, mouse_wgs_rna_tmp)
  }
  Mouse_Data_scRNA_benchmarking[[sampleID]] <- mouse_wgs_sample_ini
}

set.seed(2025)
line_num <- 0
for (iter in 1:iteration_times){
  for (sampleID in SampleIDs_Mouse){
    Mouse_Data_scRNA_benchmarking_bm <- Mouse_Data_scRNA_benchmarking[[sampleID]] %>% sample_frac(sampl
    for (tool in Tools_benchmarking_Mouse){
      line_num <- line_num + 1
      Mouse_Data_scRNA_benchmarking_bm_tool <- Mouse_Data_scRNA_benchmarking_bm %>% filter(Tool_ata ==
      total_calls <- sum(!is.na(Mouse_Data_scRNA_benchmarking_bm_tool$chrom_ata))
      total_truth <- sum(!is.na(Mouse_Data_scRNA_benchmarking_bm$chrom_wgs))
      tp <- nrow(na.omit(Mouse_Data_scRNA_benchmarking_bm_tool))
      fp <- total_calls - tp
      fn <- total_truth - tp
      sensitivity <- ifelse(total_truth > 0, tp / (tp + fn), NA)
      precision   <- ifelse(total_calls > 0, tp / (tp + fp), NA)
      f1          <- ifelse(!is.na(sensitivity) & !is.na(precision) &
                            (sensitivity + precision) > 0,
                            2 * (precision * sensitivity) / (precision + sensitivity),
                            NA)
      data_benchmarking_Mouse_scRNA_res[line_num, "Iteration"] <- iter
      data_benchmarking_Mouse_scRNA_res[line_num, "sampleID"] <- sampleID
      data_benchmarking_Mouse_scRNA_res[line_num, "Tool"] <- tool
      data_benchmarking_Mouse_scRNA_res[line_num, "Precision"] <- precision
      data_benchmarking_Mouse_scRNA_res[line_num, "Recall"] <- sensitivity
      data_benchmarking_Mouse_scRNA_res[line_num, "F1Score"] <- f1
    }
  }
}

colorAssign_1 = c(freebayes = "#FFDC91FF", SAMtools = "#7876B1FF",
                  scAllele = "#20854EFF", VarScan2 = "#0072B5FF",
                  strelka2= "#9B9898", SComatic = "#fb9a99",
```

```
                  Monopogen = "#6F99ADFF", scMutrace = "#BC3C29FF")

data_benchmarking_Mouse_scRNA_res <- data_benchmarking_Mouse_scRNA_res %>%
  mutate(Tool = fct_reorder(Tool, F1Score, .fun = mean, .na_rm = TRUE))

data_benchmarking_Mouse_scRNA_res <- na.omit(data_benchmarking_Mouse_scRNA_res)

p_Mouse_scRNA_point <- ggplot(data = data_benchmarking_Mouse_scRNA_res) +
  geom_point(mapping =  aes(x = Recall, y = Precision, color = Tool), size = 5, show.legend = FALSE) +
  scale_color_manual(values = colorAssign_1) +
  xlab("Recall") + ylab("Precision") +
  xlim(0, 0.15) +
  ylim(0, 1) +
  theme_classic()

p_Mouse_scRNA_box <- ggplot(data = data_benchmarking_Mouse_scRNA_res, aes(x = Tool, y = F1Score)) +
  stat_summary(mapping=aes(fill = Tool),fun=mean, geom = "bar", fun.args = list(mult=1),width=0.7) +
  stat_summary(fun.data=mean_sdl,fun.args = list(mult=1), geom="errorbar",width=0.2) +
  scale_fill_manual(values = colorAssign_1) +
  labs(x = "",y = "F-score")  +
  scale_x_discrete(guide = guide_axis(angle = 45)) +
  theme_classic() + guides(fill=guide_legend(title="Method"))

p_Mouse_scRNA_point
```



```
p_Mouse_scRNA_box
```

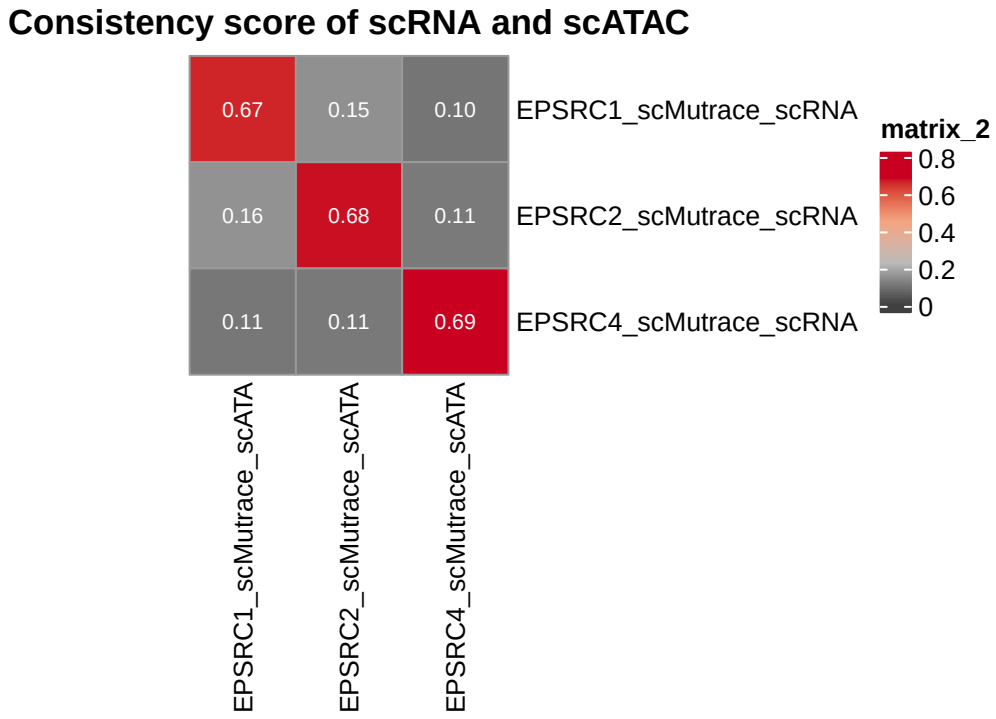



```
ggsave("../Figures/p_Mouse_scRNA_benchmarking_point.pdf", p_Mouse_scRNA_point, width = 4.47, height = 3
ggsave("../Figures/p_Mouse_scRNA_benchmarking_box.pdf", p_Mouse_scRNA_box, width = 4.47, height = 3.75)
```

9 Mouse scRNA scATA calConsistencyscore

```
SampleIDs_Mouse <- c("EPSRC1", "EPSRC2", "EPSRC4")
Mouse_Consistencyscore <- readRDS("../Data/GermlineMutation_results_Mouse_Consistencyscore.rds")

Mouse_consistencyScoreMatrix_RNA_ATA <- matrix(nrow = length(SampleIDs_Mouse), ncol = length(SampleIDs_
colnames(Mouse_consistencyScoreMatrix_RNA_ATA) <- SampleIDs_Mouse
rownames(Mouse_consistencyScoreMatrix_RNA_ATA) <- SampleIDs_Mouse

for (i in 1:nrow(Mouse_consistencyScoreMatrix_RNA_ATA)){
  sampleID_i <- SampleIDs_Mouse[i]
  rna_tmp <- Mouse_Consistencyscore[[sampleID_i]] %>% filter(sequencingType == "scRNA")
  rna_tmp$id <- paste(rna_tmp$chrom, rna_tmp$pos_start, rna_tmp$ref, sep = "_")
  rna_tmp_id <- rna_tmp$id

  for (j in 1:ncol(Mouse_consistencyScoreMatrix_RNA_ATA)){
    sampleID_j <- SampleIDs_Mouse[j]
```

```
    ata_tmp <- Mouse_Consistencyscore[[sampleID_j]] %>% filter(sequencingType == "scATA")
    ata_tmp$id <- paste(ata_tmp$chrom, ata_tmp$pos_start, ata_tmp$ref, sep = "_")
    ata_tmp_id <- ata_tmp$id

    inter_len <- length(intersect(rna_tmp_id, ata_tmp_id))
    union_len <- length(union(rna_tmp_id, ata_tmp_id))

    score_tmp <- ifelse(union_len == 0, NA, inter_len / union_len)

    Mouse_consistencyScoreMatrix_RNA_ATA[i, j] <- score_tmp
  }
}

rownames(Mouse_consistencyScoreMatrix_RNA_ATA) <- paste0(SampleIDs_Mouse, "_scMutrace_scRNA")
colnames(Mouse_consistencyScoreMatrix_RNA_ATA) <- paste0(SampleIDs_Mouse, "_scMutrace_scATA")

max_val <- max(Mouse_consistencyScoreMatrix_RNA_ATA)

col_fun_v1 <- colorRamp2(c(0, max_val * 0.33, max_val * 0.66, max_val), c("#404040", "#bababa", "#f4a58

p_consistencyScoreMatrix_RNA_ATA <- pheatmap(Mouse_consistencyScoreMatrix_RNA_ATA,
                                             cellwidth = 40, cellheight = 40,
                                             cluster_rows = F, cluster_cols = F,
                                             display_numbers = TRUE, number_color = "white",
                                             main = "Consistency score of scRNA and scATAC",
                                             color = col_fun_v1)

p_consistencyScoreMatrix_RNA_ATA
```

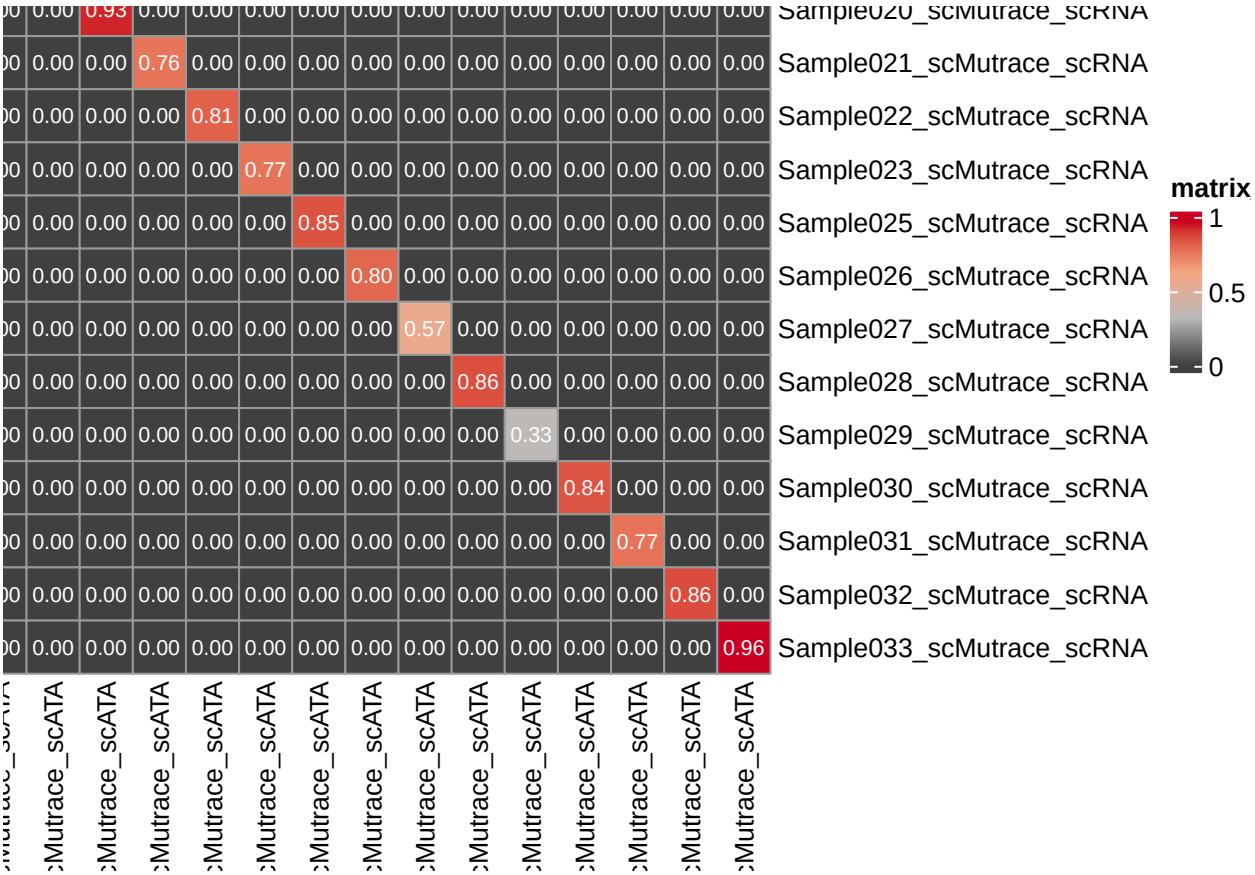



```
pdf("../Figures/p_Mouse_ConsistencyScore_scRNA_scATAC.pdf", width = 4.47, height = 3.75)
p_consistencyScoreMatrix_RNA_ATA
dev.off()
## pdf
##   2
```

10 BrainPFC

```
# common region with at least 2 cells
PFC_Mutation <- readRDS("../Data/BrainPFC_Mutation_res.rds")

PFC_sampleIDs <- c("Sample018", "Sample019", "Sample020", "Sample021", "Sample022",
                   "Sample023", "Sample025", "Sample026", "Sample027", "Sample028",
                   "Sample029", "Sample030", "Sample031", "Sample032", "Sample033")

consistencyScoreMatrix_RNA_ATA_PFC <- matrix(nrow = length(PFC_sampleIDs), ncol = length(PFC_sampleIDs))

for (i in 1:nrow(consistencyScoreMatrix_RNA_ATA_PFC)){
  # sampleID
  sample_RNA_id <- paste0(PFC_sampleIDs[i], "_RNA")
  # data
  sample_RNA_data <- PFC_Mutation[[sample_RNA_id]]
  sample_RNA_data$tagIndex <- paste0(sample_RNA_data$V1, "_",sample_RNA_data$V2)
```

```
  tagIndex_RNA <- sample_RNA_data$tagIndex
  for (j in 1:ncol(consistencyScoreMatrix_RNA_ATA_PFC)){
    sample_ATA_id <- paste0(PFC_sampleIDs[j], "_ATA")
    sample_ATA_data <- PFC_Mutation[[sample_ATA_id]]
    sample_ATA_data$tagIndex <- paste0(sample_ATA_data$V1, "_",sample_ATA_data$V2)
    tagIndex_ATA <- sample_ATA_data$tagIndex
    consistencyScoreMatrix_RNA_ATA_PFC[i, j] <- length(unique(intersect(tagIndex_RNA, tagIndex_ATA)))/le
    rm(sample_ATA_data)
  }
  rm(sample_RNA_data)
}

rownames(consistencyScoreMatrix_RNA_ATA_PFC) <- paste0(PFC_sampleIDs, "_scMutrace_scRNA")
colnames(consistencyScoreMatrix_RNA_ATA_PFC) <- paste0(PFC_sampleIDs, "_scMutrace_scATA")

max_val <- max(consistencyScoreMatrix_RNA_ATA_PFC)

col_fun_v1 <- colorRamp2(c(0, max_val * 0.33, max_val * 0.66, max_val), c("#404040", "#bababa", "#f4a582

p_consistencyScoreMatrix_RNA_ATA_brain <- pheatmap(consistencyScoreMatrix_RNA_ATA_PFC,
                                                   cellwidth = 20, cellheight = 20,
                                                   cluster_rows = F, cluster_cols = F,
                                                   display_numbers = TRUE, number_color = "white",
                                                   main = "Consistency score of scRNA and scATAC",
                                                   color = col_fun_v1)

p_consistencyScoreMatrix_RNA_ATA_brain
```




```
pdf("../Figures/p_Brain_ConsistencyScore_scRNA_scATAC.pdf", width = 8.50, height = 6.70)
p_consistencyScoreMatrix_RNA_ATA_brain
dev.off()
## pdf
##   2
```

11 sessionInfo

```
sessionInfo()
## R version 4.3.2 (2023-10-31)
## Platform: aarch64-apple-darwin20 (64-bit)
## Running under: macOS 15.6.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
```

```
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] ComplexHeatmap_2.18.0 bedtoolsr_2.30.0-6    gridExtra_2.3
##  [4] patchwork_1.2.0       circlize_0.4.16       pheatmap_1.0.12
##  [7] reshape2_1.4.4        ggplot2_3.5.1         forcats_1.0.0
## [10] readxl_1.4.3          scales_1.3.0          dplyr_1.1.4
## [13] ggsci_3.2.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] digest_0.6.37       rpart_4.1.23        lifecycle_1.0.4
##  [7] cluster_2.1.6       Cairo_1.6-2         magrittr_2.0.3
## [10] compiler_4.3.2      rlang_1.1.4         Hmisc_5.1-3
## [13] tools_4.3.2         utf8_1.2.4          yaml_2.3.10
## [16] data.table_1.16.0   knitr_1.48          labeling_0.4.3
## [19] htmlwidgets_1.6.4   plyr_1.8.9          RColorBrewer_1.1-3
## [22] withr_3.0.1         foreign_0.8-87      BiocGenerics_0.48.1
## [25] nnet_7.3-19         stats4_4.3.2        fansi_1.0.6
## [28] colorspace_2.1-1    iterators_1.0.14    cli_3.6.3
## [31] rmarkdown_2.28      crayon_1.5.3        ragg_1.3.2
## [34] generics_0.1.3      rstudioapi_0.16.0   rjson_0.2.21
## [37] stringr_1.5.1       parallel_4.3.2      cellranger_1.1.0
## [40] matrixStats_1.3.0   base64enc_0.1-3     vctrs_0.6.5
## [43] IRanges_2.36.0      GetoptLong_1.0.5    S4Vectors_0.40.2
## [46] Formula_1.2-5       htmlTable_2.4.3     clue_0.3-66
## [49] magick_2.8.7        systemfonts_1.1.0   foreach_1.5.2
## [52] glue_1.7.0          codetools_0.2-20    stringi_1.8.4
## [55] shape_1.4.6.1       gtable_0.3.5        munsell_0.5.1
## [58] tibble_3.2.1        pillar_1.9.0        htmltools_0.5.8.1
## [61] R6_2.5.1            textshaping_0.4.0   doParallel_1.0.17
## [64] evaluate_0.24.0     highr_0.11          png_0.1-8
## [67] backports_1.5.0     Rcpp_1.0.13         checkmate_2.3.2
## [70] xfun_0.47           pkgconfig_2.0.3     GlobalOptions_0.1.2
```